Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If a reaction involves only solids and liquids which of the following is true?

(A) $\Delta \mathrm{H}<\Delta \mathrm{E}$

(B) $\Delta \mathrm{H}=\Delta \mathrm{E}$

(C) $\Delta \mathrm{H}>\Delta \mathrm{E}$

(D) $\Delta \mathrm{H}=\Delta \mathrm{E}+\Delta \mathrm{nRT}$

Q.2

One mole of a non-ideal gas undergoes a change of state $(2.0\ \mathrm{atm}, 3.0\ \mathrm{L}, 95\ \mathrm{K}) \rightarrow (4.0\ \mathrm{atm},$ 5.0 L, $245\ \mathrm{K}$) with a change in internal energy, $\Delta \mathrm{U}=30.0\ \mathrm{L\ atm}$. The change in enthalpy $\Delta \mathrm{H}$ of the process in L atm is.

(A) 40.0

(B) 42.3

(C) 44.0

(D) Not defined because pressure is not constant

Q.3

If bond enthalpies of $\mathrm{N} \equiv \mathrm{N}, \mathrm{H}-\mathrm{H}$ and $\mathrm{N}-\mathrm{H}$ bonds are x_1, x_2 and x_3 respectively, $\Delta \mathrm{H}_{\mathrm{f}}^{\mathrm{o}}$ for NH_3 will be

(A) $x_1+3 x_2-6 x_3$

(B) $1/2 \mathrm{x}_1+3/2 \mathrm{x}_2-3 \mathrm{x}_3$

(C) $3 x_3-1/2 x_1-3/2 x_2$

(D) $6 x_3-x_1-3 x_2$

Q.4

Heat of neutralization of a strong acid HA and a weaker acid HB with KOH are -13.7 and $-12.7 \mathrm{k}$ cal mol^{-1}. When 1 mole of KOH was added to a mixture containing 1 mole each of HA and HB, the heat change was $-13.5 \mathrm{kcal}$. In what ratio is the base distributed between HA and HB.

(A) $3:1$

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(B) $1:3$

(C) $4:1$

(D) $1:4$

Q.5

The latent heat of vapourization of a liquid at $500\ \mathrm{K}$ and 1 atm pressure is $10.0 \mathrm{kcal/mol}$. What will be the change in internal energy $(\Delta \mathrm{U})$ of 3 moles of liquid at the same temperature

(A) $13.0 \mathrm{kcal/mol}$

(B) $-13.0 \mathrm{kcal/mol}$

(C) $27.0 \mathrm{kcal}$

(D) $-7.0 \mathrm{kcal/mol}$

Q.6

Given that:
(i) $\Delta_{\mathrm{f}} \mathrm{H}^{\circ}$ of $\mathrm{N_2O}$ is $82\ \mathrm{kJ\ mol^{-1}}$
(ii) Bond energies of $\mathrm{N \equiv N, N=N, O=O}$
and $\mathrm{N=O}$ are $946, 418, 498$ and $607\ \mathrm{kJ\ mol^{-1}}$ respectively, The resonance energy of $\mathrm{N_2O}$ is :

(A) $-88\ \mathrm{kJ}$

(B) $-66\ \mathrm{kJ}$

(C) $-62\ \mathrm{kJ}$

(D) $-44\ \mathrm{kJ}$

Q.7

The difference between the reaction enthalpy change $(\Delta_{\mathrm{r}} \mathrm{H})$ and reaction internal energy change $(\Delta_{\mathrm{r}} \mathrm{U})$ for the reaction:
$\mathrm{2C_6H_6(l) + 15O_2(\ g) \longrightarrow 12CO_2(\ g) + 6H_2O(l)}$
at $300\ \mathrm{K}$ is $\left(\mathrm{R} = 8.314\ \mathrm{J\ mol^{-1}\ K^{-1}}\right)$

(A) $0\ \mathrm{J\ mol^{-1}}$

(B) $2490\ \mathrm{J\ mol^{-1}}$

(C) $-2490\ \mathrm{J\ mol^{-1}}$

(D) $-7482\ \mathrm{J\ mol^{-1}}$

Q.8

From given following equations and $\Delta \mathrm{H}^{\circ}$ values, determine the enthalpy of reaction at $298\ \mathrm{K}$ for the

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reaction :

$$C_2H_4(g) + 6\,F_2(g) \longrightarrow 2CF_4(g) + 4HF(g)$$
$$H_2(g) + F_2(g) \longrightarrow 2HF(g);\ \Delta H_1^\circ = -537\ \text{kJ}$$
$$C(s) + 2\,F_2(g) \longrightarrow CF_4(g);\ \Delta H_2^\circ = -680\ \text{kJ}$$
$$2C(s) + 2H_2(g) \longrightarrow C_2H_4(g);\ \Delta H_3^\circ = 52\ \text{kJ}$$

(A) -1165

(B) -2486

(C) $+1165$

(D) $+2486$

Q.9

Substance $A_2\,B(g)$ can undergoes decomposition to form two set of products :

$$A_2B(g) \longrightarrow \begin{matrix} \longrightarrow A_2(g) + B(g);\ \Delta H^\circ = 40\ \text{kJ/mol} \\ \longrightarrow A(g) + AB(g);\ \Delta H^\circ = 50\ \text{kJ/mol} \end{matrix}$$

If the molar ratio of $A_2(g)$ to $A(g)$ is $5:3$ in a set of product gases, then the energy involved in the decomposition of 1 mole of $A_2\,B(g)$ is :

(A) $48.75\ \text{kJ/mol}$

(B) $43.75\ \text{kJ/mol}$

(C) $46.25\ \text{kJ/mol}$

(D) None of these

Q.10

$36\ \text{mL}$ of pure water takes 100sec to evaporate from a vessel and heater connected to an electric source which delivers 806 watt. The ΔH_{vap} of H_2O is :

(A) $40.3\ \text{kJ/mol}$

(B) $43.2\ \text{kJ/mol}$

(C) $4.03\ \text{kJ/mol}$

(D) None of these

Q.11

The enthalpy of the reaction forming PbO according to the following equation is $438\ \text{kJ}$. What heat energy (kJ) is released in formation of $22.3\ \text{g}\,PbO(s)$?

(Atomic masses : $Pb = 207, O = 16.0$)

$$2\,Pb(s) + O_2(g) \longrightarrow 2PbO(s)$$

(A) 21.9

(B) 28.7

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(C) 14.6

(D) 34.2

Q.12

The species which by definition has zero standard molar enthalpy of formation at $298\ \mathrm{K}$ is

(A) $\mathrm{Br_2(\ g)}$

(B) $\mathrm{Cl_2(\ g)}$

(C) $\mathrm{H_2O(g)}$

(D) $\mathrm{CH_4(\ g)}$

Q.13

What is $\Delta \mathrm{n_{gas}}$ for the combustion of 1 mole of benzene, when both the reactants and the products are at $298\ \mathrm{K}$?

(A) 0

(B) $1/2$

(C) $3/2$

(D) $-3/2$

Q.14

The enthalpy of neutralisation of $\mathrm{NH_4OH}$ and $\mathrm{CH_3COOH}$ is $-10.5\mathrm{kcalmol^{-1}}$ and enthalpy of neutralisation of $\mathrm{CH_3COOH}$ with strong base is $-12.5\mathrm{kcalmol^{-1}}$. The enthalpy of ionisation of $\mathrm{NH_4OH}$ will be

(A) 3.2 kcal mol $^{-1}$

(B) $2.0\mathrm{kcalmol^{-1}}$

(C) $3.0\mathrm{kcalmol^{-1}}$

(D) $4.0\mathrm{kcalmol^{-1}}$

Q.15

The enthalpy of hydrogenation of cyclohexene is $-119.5\ \mathrm{kJ\ mol^{-1}}$. If resonance energy of benzene is $-150.4\ \mathrm{kJ\ mol^{-1}}$, its enthalpy of hydrogenation would be

(A) $-208.1\ \mathrm{kg\ mol^{-1}}$

(B) $-269.9\ \mathrm{kg\ mol^{-1}}$

(C) $-358.5\ \mathrm{kg\ mol^{-1}}$

(D) $-508.9\ \mathrm{kg\ mol^{-1}}$

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Q.16

The standard enthalpies of formation at $300\ \mathrm{K}$ for $\mathrm{CCL_4(l)}, \mathrm{H_2O(g)}, \mathrm{CO_2(\ g)}$ and $\mathrm{HCl(g)}$ are $-107, -242, -394$ and $-93\ \mathrm{kJ\ mol^{-1}}$, respectively. The value of ΔU for the reaction $\mathrm{CCl_4}(l) + 2\mathrm{H_2O(g)} \rightarrow \mathrm{CO_2(\ g)} + 4\mathrm{HCl(g)}$ is

(A) $-170\ \mathrm{kJ\ mol^{-1}}$

(B) $-175\ \mathrm{kJ\ mol^{-1}}$

(C) $-182.5\ \mathrm{kJ\ mol^{-1}}$

(D) $-282.5\ \mathrm{kJ\ mol^{-1}}$

Q.17

For which one of the following equations is $\Delta_x H^\circ$ equal to $\Delta_\mathrm{f}\mathrm{H}^\circ$?

(A) $\mathrm{CH_4(\ g) + 2Cl_2(\ g) \rightarrow CH_2Cl_2(l) + 2HCI(g)}$

(B) $\mathrm{Xe(g) + 2\ F_2(\ g) \rightarrow XeF_4(\ g)}$

(C) $\mathrm{2CO(g) + O_2(\ g) \rightarrow 2CO_2(\ g)}$

(D) $\mathrm{N_2(\ g) + O_3(\ g) \rightarrow N_2O_3(\ g)}$

Q.18

The magnitude of enthalpy of neutralization of a weak acid and a strong base may be

(A) equal to $57.32\ \mathrm{kJ\ mol^{-1}}$

(B) greater than $57.32\ \mathrm{kJ\ mol^{-1}}$

(C) lesser than $57.32\ \mathrm{kJ\ mol^{-1}}$

(D) greater or lesser than $57.32\ \mathrm{kJ\ mol^{-1}}$ depending upon acid

Q.19

The enthalpy of combustion of $\mathrm{H_2(\ g)}$ to give $\mathrm{H_2O(g)}$ is $-249\ \mathrm{kJmo\Gamma^{-1}}$ and bond enthalpies of $\mathrm{H-H}$ and $\mathrm{O=O}$ are $433\ \mathrm{kJ\ mol^{-1}}$ and $492\ \mathrm{kJmo\Gamma^{-1}}$, respectively. The bond enthalpy of $\mathbf{OH}$ is

(A) $464\ \mathrm{kJ\ mol^{-1}}$

(B) $-464\ \mathrm{kJ\ mol^{-1}}$

(C) $232\ \mathrm{kJ\ mol^{-1}}$

(D) $-232\ \mathrm{kJ\ mol^{-1}}$

Q.20

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An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1. The other chamber has volume V_2 and contains ideal gas at pressure P_2 and temperature T_2. If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be

(A) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_2+P_2V_2T_1}$

(B) $\frac{P_1V_1T_1+P_2V_2T_2}{P_1V_1+P_2V_2}$

(C) $\frac{P_1V_1T_2+P_2V_2T_1}{P_1V_1+P_2V_2}$

(D) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_1+P_2V_2T_2}$

Answers & Solutions

Q.1 Answer:

$\Delta \mathrm{H}=\Delta \mathrm{E}$

Solution:

$\Delta \mathrm{H}=\Delta \mathrm{E}+\mathrm{P} \Delta \mathrm{V}$, **for solid and liquid,** $\Delta \mathrm{V}=0$
or $\Delta \mathrm{H}=\Delta \mathrm{E}+\Delta \mathrm{nRT}$, **for solids and liquids**
$\Delta \mathrm{n}=0$

Q.2 Answer:

44.0

Solution:

$\Delta \mathrm{H}=\mathrm{H}_2-\mathrm{H}_1=\left(\mathrm{E}_2+\mathrm{P}_2 \mathrm{~V}_2\right)-\left(\mathrm{E}_1+\mathrm{P}_1 \mathrm{~V}_1\right)$
$=\left(\mathrm{E}_2-\mathrm{E}_1\right)+\left(\mathrm{P}_2 \mathrm{~V}_2-\mathrm{P}_1 \mathrm{~V}_1\right)$
$=30+(4 \times 5-2 \times 3)=44 \mathrm{~L}$ **atm**

Q.3 Answer:

$1/2 \mathrm{x}_1+3/2 \mathrm{x}_2-3 \mathrm{x}_3$

Solution:

$\frac{1}{2} \mathrm{~N}_2+\frac{3}{2} \mathrm{H}_2 \longrightarrow \mathrm{NH}_3$
$\frac{1}{2} \mathrm{x}_1+\frac{3}{2} \mathrm{x}_2-3 \mathrm{x}_3$

Q.4 Answer:

$4:1$

Solution:

Let x **mole of** KOH **be neutralized by the strong acid** HA**. Then, moles neutralized by** $HB=$ $1-x$

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Hence, $-13.7 \times x+(-12.7) \times(1-x)=-13.5$

$\Rightarrow \quad x=0.8; \quad \frac{x}{1-x}=\frac{0.8}{0.2}=\frac{4}{1} 4: 1$

Q.5 Answer:

$27.0 \mathrm{kcal}$

Solution:

$3 \mathrm{H}_2 \mathrm{O}(\mathrm{l}) \rightarrow 3 \mathrm{H}_2 \mathrm{O}(\mathrm{g})$

$\Delta \mathrm{n}=3, \Delta \mathrm{E}=\Delta \mathrm{H}-\Delta \mathrm{nRT}$

$=30-3 \times \frac{2}{1000} \times 500=27 \mathrm{kcal}$

Q.6 Answer:

$-88 \mathrm{~kJ}$

Solution:

$\mathrm{N}_2(\mathrm{~g})+\frac{1}{2} \mathrm{O}_2 \rightarrow \mathrm{N}_2 \mathrm{O}(\mathrm{g})$

$\mathrm{N} \equiv \mathrm{N}(\mathrm{g})+\frac{1}{2}(\mathrm{O}=\mathrm{O}) \rightarrow \ddot{\mathrm{N}}=\overset{+}{\mathrm{N}}=\ddot{\mathrm{O}}(\mathrm{g})$

$\Delta \mathrm{H}_{\mathrm{f}}^{\mathrm{o}}=[$ **Energy required for breaking of bonds]-[Energy released for forming of bonds]**

$=\left(\Delta \mathrm{H}_{\mathrm{N} \equiv \mathrm{N}}+\frac{1}{2} \Delta \mathrm{H}_{\mathrm{O}=\mathrm{o}}-\left(\Delta \mathrm{H}_{\mathrm{N}=\mathrm{N}}+\Delta \mathrm{H}_{\mathrm{N}=0}\right)\right.$

$=\left(946+\frac{1}{2} \times 498\right)-(418+607)$

$=170 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Resonance energy $=82-170=-88 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Q.7 Answer:

$-7482 \mathrm{~J} \mathrm{~mol}^{-1}$

Solution:

$\Delta H=\Delta U+\Delta n_{\mathrm{g}} R T$

For the reaction $\Delta n_{\mathrm{g}}=12-15=-3 \Delta H-\Delta U=-3 \times 8.314 \times 300$

$=-7482 \mathrm{~J} \mathrm{~mol}^{-1}$

Q.8 Answer:

-2486

Solution:

$\Delta \mathrm{H}^{\circ}=2 \times \Delta \mathrm{H}_1^{\circ}+2 \times \Delta \mathrm{H}_2^{\circ}-\Delta \mathrm{H}_3^{\circ}=-2486 \mathrm{~kJ}$

Q.9 Answer:

$43.75 \mathrm{~kJ} / \mathrm{mol}$

Solution:

$\Delta_{\mathrm{r}} \mathrm{H}=\frac{5}{8} \times 40+\frac{3}{8} \times 50=43.75 \mathrm{~kJ} / \mathrm{mol}$

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Q.10 Answer:

$40.3 \mathrm{~kJ} / \mathrm{mol}$

Solution:

1 watt $=1 \mathrm{~J} / \mathrm{sec}$

Total heat supplied for $36 \mathrm{~mL} \mathrm{H}_2 \mathrm{O}$

$$
\begin{aligned}
& =806 \times 100 \\
& =80600 \mathrm{~J} \\
\Delta \mathrm{H}_{\mathrm{vap}} & =\frac{80600}{36} \times 18 \\
=40300 & \mathrm{~J} \text{ or } 40.3 \mathrm{~kJ} / \mathrm{mol}
\end{aligned}
$$

Q.11 Answer:

21.9

Solution:

$\mathrm{Q}=\frac{1}{2} 438 \times \frac{22.3}{223}=21.9 \mathrm{~kJ}$

Q.12 Answer:

$\mathrm{Cl}_2(\mathrm{~g})$

Solution:

The species in its elemental form has zero standard molar enthalpy of formation at $298 \mathrm{~K}$**. At** $298 \mathrm{~K}, \mathrm{Cl}_2$ **is gas while** Br_2 **is liquid.**

Q.13 Answer:

$-3 / 2$

Solution:

$\mathrm{C}_6 \mathrm{H}_6(\mathrm{l})+\frac{15}{2} \mathrm{O}_2(\mathrm{~g}) \rightarrow$
$.6 \mathrm{CO}_2(\mathrm{~g})+3 \mathrm{H}_2 \mathrm{O}$ **(l)**
$\Delta \mathrm{n}=6-15 / 2=-3 / 2$

Q.14 Answer:

$2.0 \mathrm{kcalmol}^{-1}$

Solution:

$\Delta_{\text {neu }} \mathrm{H}$ **for strong base and strong acid**
$=-13.7 \mathrm{kcaleq}^{-1}$
$\Delta \mathrm{H}_{\text {ion }}\left(\mathrm{CH}_3 \mathrm{COOH}\right)$
$=-12.5-(-13.7)=1.2$ **kcal mol** $^{-1}$
$\Delta \mathrm{H}_{\text {ion }}\left(\mathrm{NH}_4 \mathrm{OH}\right)$
$=-10.5-(-13.7)-\Delta \mathrm{H}_{\text {ion }}\left(\mathrm{CH}_3 \mathrm{COOH}\right)$

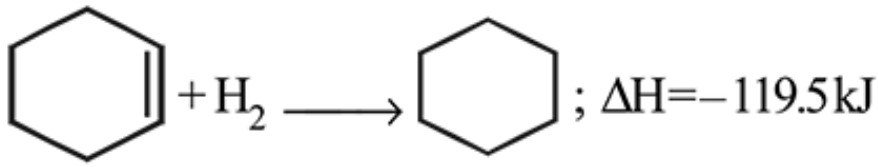
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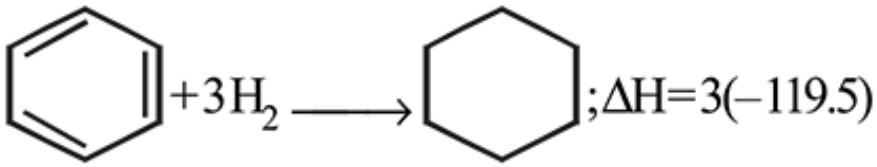
$= 13.7 - 10.5 - 1.2$
$= 2 \text{kcalmol}^{-1}$

Q.15 Answer:

$-208.1 \text{ kg mol}^{-1}$

Solution:





$= -358.5 \text{ kJ}$

The resonance energy provides extra stability to the benzene molecule so it has to be overcome for hydrogenation to take place. So $\Delta H = -358.5 - (-150.4) = -208.1 \text{ kJ}$

Q.16 Answer:

$-182.5 \text{ kJ mol}^{-1}$

Solution:

For the reaction:

$\text{CCI}_4(l) + 2\text{H}_2\text{O(g)} \rightarrow \text{CO}_2(\text{ g}) + 4\text{HCI(g)}$, **we have**

$\Delta_\text{r}\text{H} = \Delta\text{HH}\left(\text{CO}_2,\ \text{g}\right) + 4\Delta\text{fH}(\text{HCI, g}) - \Delta_\text{f}\text{H}\left(\text{CCI}_4, l\right) - 2\Delta_\text{f}\text{H}\left(\text{H}_2\text{O, g}\right)$

$= (-394 - 4 \times 93 + 107 + 2 \times 242)\text{kJmol}^{-1} = -175 \text{ kJ mol}^{-1}$

$\Delta_r\text{U} = \Delta_\text{r}\text{H} - (\Delta\text{V}_\text{g})\,\text{RT} = -175 - (3)$

$\left(8.314 \times 10^{-3} \text{ kJ K}^{-1} \text{ mol}^{-1}\right)(300 \text{ K}) = -182.5 \text{ kJ mol}^{-1}$

Q.17 Answer:

$\text{Xe(g)} + 2\ \text{F}_2(\ \text{g}) \rightarrow \text{XeF}_4(\ \text{g})$

Solution:

The reaction given in choice (b) represents either $\Delta_x\text{H}^\circ$ **or** ΔH°

Q.18 Answer:

lesser than $57.32 \text{ kJ mol}^{-1}$

Solution:

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A portion of heat is utilized in the ionization of weak acid. Hence, the magnitude of enthalpy of neutralization is less than $57.32 \mathrm{~kJ~mol}^{-1}$.

Q.19 Answer:

$464 \mathrm{~kJ~mol}^{-1}$

Solution:

For the equation

$\mathrm{H}_2(\mathrm{~g})+\frac{1}{2} \mathrm{O}_2(\mathrm{~g}) \rightarrow \mathrm{H}_2 \mathrm{O}(\mathrm{g})$

$\Delta \mathrm{H}=-249 \mathrm{~kJ~mol}^{-1}$

Thus $\varepsilon(\mathrm{H}-\mathrm{H})+\frac{1}{2} \mathrm{e}(\mathrm{O}=\mathrm{O})-2 \varepsilon(\mathrm{O}-\mathrm{H})=-249 \mathrm{~kJ~mol}^{-1}$

$=\left[433+\frac{1}{2}(492)\right] \mathbf{kJ}\,\mathrm{mol}^{-1}-2 \varepsilon(\mathrm{O}-\mathbf{H})=-249 \mathrm{~kJmol}\Gamma^{-1}$

Hence $\varepsilon(\mathrm{O}-\mathrm{H})=\frac{1}{2}\left[+249+433+\frac{1}{2}(492)\right] \mathrm{kJmol}^{-1}=464 \mathrm{~kJ~mol}^{-1}$

Q.20 Answer:

$\frac{T_1 T_2\left(P_1 V_1+P_2 V_2\right)}{P_1 V_1 T_2+P_2 V_2 T_1}$

Solution:

Here $Q=0$ **and** $W=0$**. Therefore from first law of thermodynamics** $\Delta U=Q+W=0$

$\therefore$ **Internal energy of the system with partition = Internal energy of the system without partition.**

$n_1 C_v T_1+n_2 C_v T_2=\left(n_1+n_2\right) C_v T$

$\therefore T=\frac{n_1 T_1+n_2 T_2}{n_1+n_2}$

But $n_1=\frac{P_1 V_1}{R T_1}$ **and** $n_2=\frac{P_2 V_2}{R T_2}$

$\therefore T=\frac{\frac{P_1 V_1}{R T_1} \times T_1+\frac{P_2 V_2}{R T_2} \times T_2}{\frac{P_1 V_1}{R T_1}+\frac{P_2 V_2}{R T_2}}$

$=\frac{T_1 T_2\left(P_1 V_1+P_2 V_2\right)}{P_1 V_1 T_2+P_2 V_2 T_1}$

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